The University of Melbourne

Semester 2 Examination —November, 2010

Department of Mathematics and Statistics

MAST30027 (620-330) Modern Applied Statistics

Exam duration: three hours
Reading time: fifteen minutes
This paper has 8 pages.

Authorised materials:
Hand-held electronic calculators may be used.

Instructions to invigilators:
A 24-page script book shall be supplied to each student.

Instructions to students:
There are 11 questions. All questions may be attempted.
The number of marks for each question is indicated after the question.
The total number of marks available is 100.

This paper may be released to the Library

1. Consider a random sample $X_1, \cdots, X_n$ from a Weibull distribution WEI($\theta^{\frac{1}{3}}, 3$) which has the following *pdf*

$$f(x|\theta) = \frac{3}{\theta} x^2 e^{-\frac{x^3}{\theta}}, \quad x > 0; \ \theta > 0.$$

Note the following results can be quoted without proof:

(i) $X_i^3 \stackrel{d}{=} \text{EXP}(\theta)$ and $E(X_i^3) = \theta, \quad i = 1, \cdots, n.$

(ii) The minimum variance (i.e. Cramer-Rao lower bound) of an unbiased estimator T of $\tau(\theta)$ is given by the information inequality

$$\text{Var}(T) \geq \frac{[\tau'(\theta)]^2}{-nE\left[\frac{d^2 \ln f(X|\theta)}{d\theta^2}\right]}.$$

Now complete the following tasks:

(a) Specify the joint *pdf* of $X_1, \cdots, X_n$.

(b) Find the score function $U(\theta) = \frac{d \ln L(\theta)}{d\theta}$ and the MLE of θ based on the sample.

(c) Show that the MLE of θ, denoted as $\hat{\theta}$, is unbiased for θ.

(d) Show that the MLE $\hat{\theta}$ is a UMVUE (uniformly minimum variance unbiased estimator).

[2 + 2 + 1 + 4 = 9 marks]

2. The following results can be used for this question:

(i) **Lehmann-Scheffe Theorem**: Suppose $\mathbf{S} = (S_1, \cdots, S_k)$ is jointly sufficient and complete for $\boldsymbol{\theta}$. If $T^* = t^*(\mathbf{S})$ is an unbiased estimator of $\tau(\boldsymbol{\theta})$, then T^* is a UMVUE of $\tau(\boldsymbol{\theta})$.

(ii) A *pdf* is said to be a member of the **regular exponential class (REC)** if it has the following form

$$f(x|\boldsymbol{\theta}) = c(\boldsymbol{\theta}) h(x) \exp\left(\sum_{j=1}^{q} \eta_j(\boldsymbol{\theta}) t_j(x)\right), \quad x \in A$$

and 0 elsewhere, where $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_q)$ has q independent parameters; and provided that the following conditions are satisfied:

1° $A = \{x : f(x|\boldsymbol{\theta}) > 0\}$ does not depend on $\boldsymbol{\theta}$;

2° $\eta_j(\boldsymbol{\theta})$'s are nontrivial, functionally independent and continuous functions of $\boldsymbol{\theta}$;

3° $t_j(x)$'s are nontrivial and their derivatives are linearly independent.

(iii) The *mgf* (moment generating function) for Poisson distribution POI(θ) is $M(t) = e^{\theta(e^t - 1)}$.

Consider a random sample $X_1, \cdots, X_n$ from a Poisson distribution POI(θ) which has the *pdf*

$$f(x|\theta) = \frac{\theta^x}{x!} e^{-\theta}, \quad x = 0, 1, 2, \cdots; \quad \theta > 0.$$

(a) Show that POI(θ) belongs to REC. In particular, specify A, $c(\boldsymbol{\theta})$, $h(x)$, $\eta_j(\boldsymbol{\theta})$ and $t_j(x)$ for the given pdf.

(b) Find a complete and sufficient statistic for θ based on the random sample $X_1, \cdots, X_n$.

(c) Let $\lambda = P(X_1 = 0)$. Show that $\hat{\lambda} = \left(1 - \frac{1}{n}\right)^{\sum_{i=1}^n X_i}$ is an unbiased estimator of λ.

(d) Is $\hat{\lambda}$ a UMVUE of λ? Why or why not?

[2 + 2 + 3 + 3 = 10 marks]

3. Consider a GLM (generalised linear model) for a random sample $Y_1, \cdots, Y_n$ which is to be observed from a gamma distribution. Here each Y_i has the *pdf*

$$f(y_i|\mu_i) = \frac{1}{\Gamma(\kappa_i)} \left(\frac{\kappa_i y_i}{\mu_i}\right)^{\kappa_i} \exp\left(-\frac{\kappa_i y_i}{\mu_i}\right) y_i^{-1}, \quad y_i > 0,\ \mu_i > 0,\ \kappa_i > 0; \quad i = 1, \cdots, n;$$

where κ_i's are known constants (the prior weights), $\Gamma(\kappa_i)$ is a gamma function, and $\mu_i = E(Y_i)$. Suppose the systematic component of the GLM is $\eta_i = \alpha + \beta x_i$, with (α, β) being the parameters of interest and x_i's being the covariate observations; and the link is $\eta_i = \mu_i^{-1}$, $i = 1, \cdots, n$.

(a) Find the log-likelihood function $\ell(\alpha, \beta) = \ln L(\alpha, \beta)$.

(b) Find $\frac{\partial \ln L}{\partial \alpha}$ and $\frac{\partial \ln L}{\partial \beta}$ which specify the score function $U(\alpha, \beta)$.

(c) Find $\frac{\partial^2 \ln L}{\partial \alpha^2}$, $\frac{\partial^2 \ln L}{\partial \beta^2}$ and $\frac{\partial^2 \ln L}{\partial \alpha \partial \beta}$. Then list the Fisher information matrix $I(\alpha, \beta)$.

(d) Given that $(\alpha^{(j)}, \beta^{(j)})$ is the current estimate in the process of finding the MLE of (α, β), we can update it to $(\alpha^{(j+1)}, \beta^{(j+1)})$ by the Newton-Raphson or Fisher scoring procedure. Specify the formula used in this update.

[3 + 2 + 4 + 2 = 11 marks]

4. Consider a random sample $T_1, \cdots, T_n$ from an exponential distribution which has the *pdf*

$$f(t|\lambda) = \lambda e^{-\lambda t}, \quad t > 0; \quad \lambda > 0.$$

Suppose we have observed $T_1 = y_1, \cdots, T_m = y_m$ and $T_{m+1} > c_{m+1}, \cdots, T_n > c_n$, where m is given, $m < n$, and $y_1, \cdots, y_m$ and $c_{m+1} \cdots, c_n$ are given numerical values. This implies that $T_1, \cdots, T_m$ are completely observed whereas $T_{m+1}, \cdots, T_n$ are partially observed in that they are right-censored. We want to use an EM algorithm to find the MLE of λ.

(a) Find the complete-data log-likelihood function $\ell(\lambda) = \ln L(\lambda)$.

(b) In the E-step, we calculate

$$Q(\lambda, \lambda^{(k)}) = E[\ln L(\lambda) \mid T_1 = y_1, \cdots, T_m = y_m, T_{m+1} > c_{m+1}, \cdots, T_n > c_n; \lambda^{(k)}]$$

where $\lambda^{(k)}$ is the current estimate of λ. Show that

$$Q(\lambda, \lambda^{(k)}) = n \ln \lambda - \lambda \sum_{i=1}^{m} y_i - \lambda \sum_{i=m+1}^{n} c_i - \frac{(n-m)\lambda}{\lambda^{(k)}}.$$

(*Note*: The result $E(T|T > c) = c + \frac{1}{\lambda}$ for the exponential random variable T considered here can be used without proof.)

(c) In the M-step, we maximise $Q(\lambda, \lambda^{(k)})$ with respect to λ to find an update $\lambda^{(k+1)}$ from $\lambda^{(k)}$. Show that

$$\lambda^{(k+1)} = \frac{n}{\sum_{i=1}^{m} y_i + \sum_{i=m+1}^{n} c_i + \frac{n-m}{\lambda^{(k)}}}.$$

[3 + 3 + 2 = 8 marks]

5. Let $X_1, \cdots, X_n$ be a random sample from binomial distribution with $X_i \stackrel{d}{=} \text{BIN}(m_i, \theta)$ and m_i being given. Thus the *pdf* of X_i is

$$f(x_i|\theta) = \binom{m_i}{x_i} \theta^{x_i}(1-\theta)^{m_i - x_i}, \quad x_i = 0, 1, \cdots, m_i; \quad i = 1, \cdots, n.$$

Suppose θ has a BETA(a, b) prior distribution with the *prior pdf* being

$$p(\theta|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1}; \quad a > 0, \ b > 0 \text{ being given}; \quad \text{also } E(\theta) = \tfrac{a}{a+b}.$$

(a) Find the posterior pdf of θ given $X_1, \cdots, X_n$.

(b) Find the Bayes estimator of θ when the squared loss function is used.

[5 + 3 = 8 marks]

6. A random number X is to be generated by the following acceptance-rejection (A-R) sampling algorithm:

1° Generate $U = u$ from Unif(0, 1) and $V = v$ from Unif(0, 1) independently.

2° If $u + v < 1$, deliver $X = \frac{1}{u^2}$; otherwise, go to 1°.

The *cdf* (cumulative distribution function) of X can be found using the following result:

$$P(X \le x) = P(\frac{1}{U^2} \le x \mid U + V < 1), \quad \text{for a given } x.$$

(a) Show that the pdf of X is $f(x) = \frac{\sqrt{x}-1}{x^2}, \quad x > 1.$

(b) Let N be the number of times steps 1° and 2° need be repeated before an X value is generated. What is the distribution of N and what is the mean $E(N)$?

[5 + 3 = 8 marks]

7. Consider a random sample $X_1, \cdots, X_n$ from GAMMA$(\theta, 3)$ distribution which has the pdf

$$f(x|\theta) = \frac{1}{2\theta^3} x^2 e^{-\frac{x}{\theta}}; \quad x > 0, \ \theta > 0.$$

It can be shown that θ is a scale parameter and $\hat{\theta} = \frac{1}{3}\bar{X}_n = \frac{1}{3n}\sum_{i=1}^n X_i$ is the MLE of θ. It can also be shown that $Q = \frac{2n\bar{X}_n}{\theta} \stackrel{d}{=} \chi^2(6n)$ and Q is a pivotal quantity.

(a) Explain what is called a pivotal quantity.

(b) Derive a 95% C.I. (confidence interval) formula for θ based on Q.

(c) Based on a sample of 50 observations from GAMMA$(\theta, 3)$, 1999 nonparametric bootstrap samples are generated to estimate se$(\hat{\theta})$ and construct C.I. for θ. The following R output is obtained where x stores the 50 generated GAMMA$(\theta, 3)$ observations:

```
> library(boot)
> scale.bt=function(x,i){scale=mean(x[i])/3
        n=length(x)
        v.scale=(n-1)*var(x[i])/(9*n^2)
        c(scale,v.scale)}

> set.seed(12345)
> boot7=boot(data=x,statistic=scale.bt,R=1999)
> boot7
```

```
ORDINARY NONPARAMETRIC BOOTSTRAP

Call:
boot(data = x, statistic = scale.bt, R = 1999)

Bootstrap Statistics :
     original       bias    std. error
t1* 5.7235485  0.011215029  0.47360035
t2* 0.2323829 -0.004787773  0.03778185

> boot.ci(boot7)

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 1999 bootstrap replicates

CALL :
boot.ci(boot.out = boot7)

Intervals :
Level      Normal              Basic             Studentized
95%   ( 4.784,  6.641 )   ( 4.754,  6.646 )   ( 4.816,  6.729 )

Level     Percentile            BCa
95%   ( 4.801,  6.693 )   ( 4.771,  6.648 )
Calculations and Intervals on Original Scale

> qchisq(0.975,300); qchisq(0.025,300)

[1] 349.8745
[1] 253.9123
```

i. From the R output find the value of $\hat{\theta}$, the bootstrap estimate of $\text{se}(\hat{\theta})$, and the 95% BCa C.I. of θ. Explain how the bootstrap estimate of $\text{se}(\hat{\theta})$ is calculated. What do B, C and a in BCa stand for?

ii. Calculate the 95% C.I. for θ using the formula obtained in (b) and the χ^2 percentile values in the R output.

[1 + 3 + 4 + 1 = 9 marks]

8. **Neyman-Pearson Lemma:** Suppose random variables $X_1, \cdots, X_n$ have the joint pdf $f(\mathbf{x}_n|\theta)$ with $\mathbf{x}_n = (x_1, \cdots, x_n)^T$. Let $\lambda(\mathbf{x}_n) = \dfrac{f(\mathbf{x}_n|\theta_1)}{f(\mathbf{x}_n|\theta_0)}$ be the likelihood ratio test statistic, and $C^* = \{\mathbf{x}_n : \lambda(\mathbf{x}_n) \geq k\}$ be a critical region for testing

$$H_0\colon\ \theta = \theta_0 \quad \text{vs.} \quad H_a\colon\ \theta = \theta_1,$$

where k is such that $P(\mathbf{X}_n \in C^* \mid \theta_0) = \alpha$. Then C^* is the most powerful critical region of size α for the test. Namely, for any critical region C such that $P(\mathbf{X}_n \in C|\theta_0) \leq \alpha$,

$$\pi_{C^*}(\theta_1) = P(\mathbf{X}_n \in C^* \mid \theta_1) \geq P(\mathbf{X}_n \in C \mid \theta_1) = \pi_C(\theta_1).$$

Question: Consider a random sample $X_1, \cdots, X_n$ from an exponential distribution EXP(θ) which has the *pdf* $f(x|\theta) = \frac{1}{\theta}e^{-\frac{x}{\theta}}$; $x > 0, \theta > 0$. It can be shown that $\hat{\theta} = \bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ is the MLE of θ.

(a) Determine an MP (most powerful) critical region for testing $H_0\colon\ \theta = \theta_0$ vs. $H_a\colon\ \theta = \theta_1$, where $\theta_1 < \theta_0$, at significance level α. Simplify the critical region as much as you can.

(b) Explain whether the critical region found in (a) is UMP (uniformly most powerful) of size α for testing $H_0\colon\ \theta \geq \theta_0$ vs. $H_a\colon\ \theta < \theta_0$.

(c) Derive a GLR (generalised likelihood ratio) test for testing $H_0\colon\ \theta = \theta_0$ vs. $H_a\colon\ \theta \neq \theta_0$. Simplify the derived critical region as much as you can.

[3 + 3 + 3 = 9 marks]

9. People have two ways to measure their blood pressures in order to determine their hypertension status. The first way is through an automated computer device that can be self-administered. The second way is through a trained observer. In order to compare the computer device with the standard method of using a trained observer, 20 individuals were recruited and their hypertension statuses were assessed by both the computer device and a trained observer. Hypertension status is defined as either hypertensive (+) if either systolic blood pressure ≥ 160 or diastolic blood pressure ≥ 95, or normotensive (−) otherwise. The observed results are summarized in the following table:

Computer device	Trained observer +	Trained observer −
+	3	7
−	1	9

Let p_C be an individual's probability that would be detected by the computer device as being hypertensive and p_T be that detected by the trained observer. Consider using McNemar's test to test H_0: $p_C = p_T$ vs. H_a: $p_C \neq p_T$ at significance level 0.05.

(a) Let Y_{21} denote the top-right observation in the table. What is the distribution of Y_{21} when H_0 is true? Find the exact p-value of the test using McNemar's binomial test statistic.

(b) The following is the R output from the McNemar's χ^2 test. Explain how the test statistic value $\chi^2 = 3.125$ is obtained.

```
> x=matrix(c(3,1,7,9),2,2)
> mcnemar.test(x, correct=T)

        McNemar's Chi-squared test with continuity correction

data:  x
McNemar's chi-squared = 3.125, df = 1, p-value = 0.0771
```

[3 + 3 = 6 marks]

10. Hastie and Tibshirani (1990, p.282) described a study to determine risk factors for kyphosis, severe forward flexion of the spine following corrective spinal surgery. The following table lists the age in months (x) for 18 subjects for whom kyphosis were present ($y = 1$), and for 22 subjects for whom kyphosis were absent ($y = 0$).

Kyphosis presence (y):	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Age in months (x):	12 15 42 52 59 73 82 91 96 105 114 120 121 128 130 139 139 157
Kyphosis presence (y):	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Age in months (x):	1 1 2 8 11 18 22 31 37 61 72 81 97 112 118 127 131 140 151 159 177 206

A logistic model $\ln \frac{\theta}{1-\theta} = \alpha + \beta_1 x + \beta_2 x^2$, where θ is the probability of kyphosis being present for a subject, is fitted to the data. The relevant results are provided in the following R output.

```
> kyph.3=glm(y~x+I(x^2), family=binomial)
> anova(kyph.3, test="Chisq")

Analysis of Deviance Table

Model: binomial, link: logit

Response: y

Terms added sequentially (first to last)

       Df Deviance Resid. Df Resid. Dev P(>|Chi|)
NULL                      39     55.051
x       1   0.5469        38     54.504   0.45959
I(x^2)  1   6.2762        37     48.228   0.01224
```

```
> summary(kyph.3)$coef

                Estimate   Std. Error   z value   Pr(>|z|)
(Intercept) -2.0460000000 0.9943478097 -2.057886 0.03960106
x            0.0600000000 0.0267808473  2.241893 0.02496829
I(x^2)      -0.0003280000 0.0001564095 -2.096700 0.03602012

> summary(kyph.3)$cov.unscaled

             (Intercept)             x        I(x^2)
(Intercept)  0.9887000000 -2.220200e-02  1.074100e-04
x           -0.0222000000  7.172100e-04 -4.039600e-06
I(x^2)       0.0001000000 -4.039600e-06  2.446400e-08
```

(a) Use both Wald test and the (generalised) likelihood ratio test to assess the significance of x^2 based on model `kyph.3`. Report the test statistic value and the p-value for each test.

(b) Express the odds of a 120-month old subject having kyphosis present in terms of α, β_1 and β_2. Then calculate the fitted value of this odds.

(c) Express the log odds ratio of a 132-month old subject having kyphosis present against a 120-month old subject having kyphosis present, in terms of β_1 and β_2.

(d) Estimate the log odds ratio derived in (c). Then find an approximate 95% C.I. for this log odds ratio, and an approximate 95% for the odds ratio.

[3 + 2 + 1 + 5 = 11 marks]

11. The following data were published in a paper by Radelet (1981), *Amer Sociol. Rev. 918–927*, who studied the effect of racial characteristics on whether individuals convicted of homicide receive the death penalty. The 326 subjects were defendants in homicide cases in 20 Florida counties during 1976 – 1977.

Defendant's Race	Victim's Race	Death Penalty	
		Yes	No
White	White	19	132
	Black	0	9
Black	White	11	52
	Black	6	97

The data constitute a $2 \times 2 \times 2$ table where defendants' race is the stratification factor.

(a) A Mantel-Haenszel test is performed returning the following R output. Explain what this test is testing for and what the conclusion is at the significance level 0.05.

```
> x11=array(0,c(2,2,2))
> x11[,,1]=c(19,0,132,9)
> x11[,,2]=c(11,6,52,97)
> mantelhaen.test(x11,correct=T)

        Mantel-Haenszel chi-squared test with continuity correction

data:  x11
Mantel-Haenszel X-squared = 5.8062, df = 1, p-value = 0.01597
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:  1.397771 11.381078
sample estimates:
common odds ratio
             3.988502
```

(b) The data in the table are rearranged so that a log-linear model `mod1` can be fitted, with the results being given in the following R output. Give the mathematical formula (of the form $\ln(\lambda) = \cdots$) for model `mod1`.

```
> murder.dat=data.frame(freq=c(19,0,11,6,132,9,52,97))
> murder.dat$D=factor(rep(c("W","B"), times=2, each=2))
> murder.dat$V=factor(rep(c("W","B"), times=4))
> murder.dat$P=factor(rep(c("Y","N"), each=4))
> murder.dat
  freq D V P
1   19 W W Y
2    0 W B Y
3   11 B W Y
4    6 B B Y
5  132 W W N
6    9 W B N
7   52 B W N
8   97 B B N
> mod1=glm(freq~D+V+P+D:V+D:P+V:P, family=poisson, data=murder.dat)
> anova(mod1, test="Chisq")
Analysis of Deviance Table

Model: poisson, link: log

Response: freq

Terms added sequentially (first to last)


     Df Deviance Resid. Df Resid. Dev P(>|Chi|)
NULL                     7     395.92
D     1    0.110         6     395.80  0.739649
V     1   32.456         5     363.35 1.219e-08
P     1  225.419         4     137.93 < 2.2e-16
D:V   1  129.798         3       8.13 < 2.2e-16
D:P   1    0.221         2       7.91  0.637937
V:P   1    7.209         1       0.70  0.007252
> 1-pchisq(0.70,1)
[1] 0.4027837
> 1-pchisq(7.91,2)
[1] 0.01915867
```

(c) What assumption is made for the Mantel-Haenszel test in (a) to be valid, in regard to the association between the victims' race and the defendants' sentence? Test this assumption using the R output in (b) and draw the conclusion at the significance level 0.05.

(d) Test the adequacy of model `D + V + P + D:V + D:P`, at significance level 0.05, using the R output in (b). Explain the result in terms of the association among defendants' race, victims' race and defendants' sentence. How is the result related to that of (a)?

[3 + 1 + 3 + 4 = 11 marks]

Total marks = 100

END OF EXAMINATION